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- 1) What are the different data types available in java
Explain with example.

Data types

Int → Integer (i.e., 1 to 20...)

char → Single char (ex: 'A', 'B', 'c')

float → with floating values (1.256f, 1.276f)

double → with double value (1.37d, 1.38d)

String → collection of characters ('Manasa')

Boolean → Boolean executes as (true or false)

byte → With byte like numbers

Example 3 (Int)

package practice

public class sample

{
 public static void main (String [] args)

{
 int a = 10; system.out.println ("a");
 int b = 20; } system.out.println ("b");
 system.out.println ("int a + int b");

}

}

Example char

{

char a = 'M';

char b = 'N'; char c = M+N;

system.out.println (char c);

}

Example float

```
{
    float a = 1.23f;
    float b = 1.33f;
    float c = 1.44f;
    system.out.println("The float sum of float values of a and b is: " + a + b);
    system.out.println("The float value of c is: " + c);
}
```

Example double

```
{
    double a = 1.55d;
    double b = 1.57d;
    double c = 1.56d;
    system.out.println("The double value of a is: " + a);
    system.out.println("The sum value of double a and b);
}
```

Example String

```
{
    String name1 = "Manasa";
    String name2 = "Sai";
    system.out.println("The name is: " + name);
    system.out.println("The name1 is: " + " " + name1 + "/n " +
        "The name 2 is: " + " " + name2);
}
```

Boolean

package

public practice

public class Sample

```
{
    public class void main (String [] args)
```

```
{ boolean = true or false;
```

```
    a = 10;
```

```
    b = 5;
```

```

if (a > b)
{
    System.out.println (True);
}
else
{
    System.out.println (false);
}
}
}

```

4) write a program to print only odd number between 1-20.

```

Package
public practice
public class OddNumber
{
    public class void main (String [] args)
    {
        int int i = 0
        if (int i < 20)
        for (int i = 0; i % 2 != 0; i++)
        {
            System.out.println ("The odd Number are: "+i);
        }
    }
}
}

```

3) Explain about looping statements with syntax and examples.

Looping: * Looping are the statements that will be executed until the certain condition becomes true.

* If the condition is true then the output of the code will be executed.

If the condition is false, it will be looped continuously.

Types of Looping:

i) for loop

ii) while loop

iii) do while loop.

iv) for each loop

Syntax

for loop:

for (Initialization ; condition ; Increment (or) decrement)

{

code to be executed if it is true;

}

while loop

Initialization;

while //condition;

{

code to be executed, if it is true

}

//Increment or decrement;

do while:

In do while loop the statement will be executed once whether it is true or false, then the condition will be checked, if it is true it is executed.

do

// Initialization

do

{

statement to be executed

}

while // condition

Example

for loop.

package

public practice

public class Even numbers

{

public static void main (String [] args)

{ int i = 0;

if (int i < 20)

{

for (int i = 0; i % 2 == 0; i++)

{

System.out.println (" The even numbers are:"
+ i);

}

}

}

}

while loop

Package practice

public class Even number

{
 ^{static} void main (String [] arg)

{

~~int i = 0;~~

int i = 0;

while

{

i % 2 == 0;

}

i++ ;

}

}

2. Explain different conditional statement with syntax and examples.

The conditional statement are used to check wheather the condition satisfy or not.

Types of conditional statement

* if

else

* else if

* ~~if~~ if - else - if ladder.

Syntax for conditional statement

if (condition)

{

// condition ~~true~~ that will be executed as expected
// checks the condition
}

else

{

if - else syntax

if (condition)

{

// condition is true that will be executed
// checks the condition

}

else

{

print (false) or any other statement that is
mentioned

}

if - else - if syntax

if (condition)

{

// checks the condition

}

else

{

(print other statement)

}

elif

{

print the statement

5

Examples for conditional statement

```
if (a > b)
{
```

```
    system.out.println("a is greater than b: " +
                        a > b)
```

```
}
```

```
if - else
```

```
if (a > b)
```

```
{
```

```
    system.out.println("a is greater than b: " +
                        a > b)
```

```
}
else
```

```
{
```

```
    system.out.println("a is gr lesser than b: " +
                        a > b)
```

```
}
```

```
if - else - if
```

```
if
```

```
if
```

```
    int manager salary = 90,000;
```

```
    int Team leader salary = 80,000;
```

```
    int Employee salary = 50,000;
```

```
    if (manager salary >= 9085,000)
```

```
    {
        system.out.println("The manager salary is: ")
```

```
        manager salary;
```

```
    }
```

```
    else if (Team leader >= 80,000)
```



```
{  
    System.out.println("The Team Leader Salary is:"  
                        + Team Leader);
```

```
}
```

```
else
```

```
{  
    System.out.println("The employee salary");
```

```
}
```
